

Practical-5

AIM: Study and Configuration of Wireless LAN (WLAN) using Cisco Packet Tracer.

➤ REQUIREMENTS

1. Any Computer or Laptop
2. Computer system with minimum 4 GB RAM
3. Cisco Packet Tracer software
4. Windows Operating System

➤ THEORY

A **Wireless Local Area Network (WLAN)** is a wireless distribution method for two or more devices that use high-frequency radio waves to communicate, often including an access point connecting to a wired network.

Main components of WLAN:

- **Access Point (AP):** A networking hardware device that allows other Wi-Fi devices to connect to a wired infrastructure. It acts as a bridge between the wired and wireless networks.
- **SSID (Service Set Identifier):** The primary network name associated with an 802.11 wireless local area network. Devices must match this SSID to join the network.
- **Wireless Modules:** In standard network simulations, end devices like desktop PCs and printers may lack built-in Wi-Fi. They require physical network interface card (NIC) replacements to gain wireless capabilities.

Advantages of WLAN:

- Mobility and flexibility
- Easy installation
- Reduced wiring cost
- Scalability

WLAN security can be implemented using **WEP, WPA, WPA2** passwords.



➤ PROCEDURE

- **Step 1: Open Cisco Packet Tracer**

1. Launch **Cisco Packet Tracer**.
2. Select **File** → **New** to create a new workspace.

- **Step 2: Add Network Devices**

From the device panel, add the following devices to the workspace.

Network Devices:

- Router **1841**
- Switch **2960-24TT**
- **Access Point**

End Devices:

- PC
- Laptop
- Smartphone
- Tablet
- Printer

- **Step 3: Connect the Wired Devices**

Use **Copper Straight-Through Cable**.

Make the following connections:

Router **Fa0/0** → Switch **Fa0/1**

Switch **Fa0/2** → Access Point **Ethernet Port**

In this configuration:

- The **router** provides **network gateway**
- The **switch** connects **network devices**
- The **access point** provides **wireless connectivity**

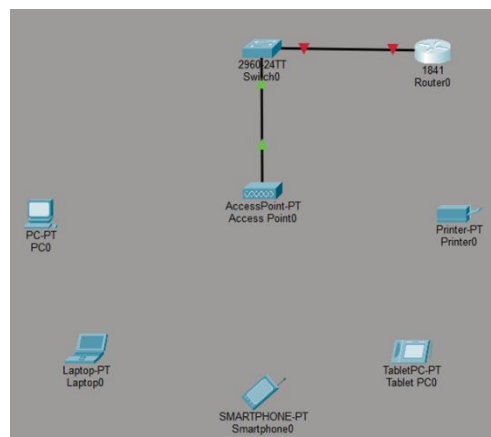


Figure 1: WLAN Network Topology in Cisco Packet Tracer



- **Step 4: Configure the Router**

- Click on the **1841 Router** and navigate to the **Config** tab.
- On the left-hand menu under **Interfaces**, select **FastEthernet0/0**.
- In the **IP Address** field, enter 192.168.0.1.
- In the **Subnet Mask** field, enter 255.255.255.0.
- In the top right corner, check the box for **Port Status** to **On**.
- This IP address will act as the **default gateway for all devices**.

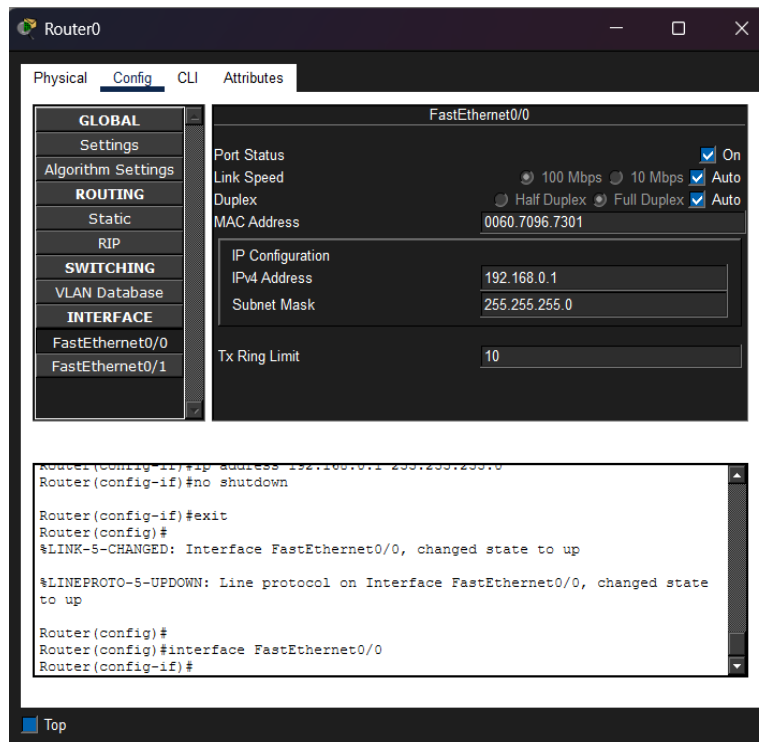


Figure 2: Router IP Address Configuration

- **Step 5: Configure the Access Point**

- Click on the **Access Point** and navigate to the **Config** tab.
- On the left-hand menu under the **Interface** section, select **Port 1**.
- Ensure **Port Status** is checked **On**.
- In the **SSID** field, enter **LabWLAN**.
- Under the **Authentication** section, select the radio button for **WEP**.
- In the **WEP Key** field, enter the 10-digit hexadecimal key: **1234567890**.
- Ensure the **Encryption Type** is set to **40/64-Bits (10 Hex digits)**

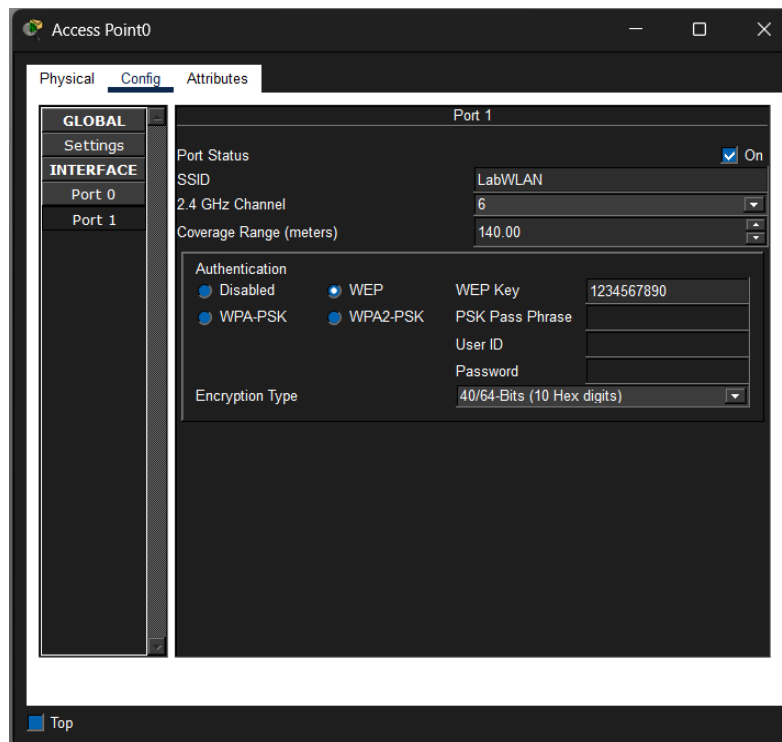


Figure 3: Ping Test Showing Successful Connectivity

- **Step 6: Configure IP Addresses of Devices**

Assign static IP addresses to all devices to ensure they can communicate within the **LabWLAN** network. Each device must use the Router's IP as its gateway to access the rest of the network.

1. Click on the specific **End Device** (PC, Laptop, Smartphone, Printer, or Tablet).
2. Navigate to the **Desktop** tab and select **IP Configuration**.
3. Ensure **Static** is selected and enter the following details based on the device:

- **Router:** IP Address 192.168.0.1, Subnet Mask 255.255.255.0
- **PC:** IP Address 192.168.0.2, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1
- **Laptop:** IP Address 192.168.0.3, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1
- **Smartphone:** IP Address 192.168.0.4, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1
- **Printer:** IP Address 192.168.0.5, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1
- **Tablet:** IP Address 192.168.0.6, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1



- **Step 7: Connect Wireless Devices**

For the PC, Laptop, Smartphone, Tablet, and Printer, follow these steps to establish a secure connection using the **Config** tab :

1. Click on the device to open its settings window.
2. Navigate to the **Config** tab and select **Wireless0** from the left-hand interface menu.
3. In the **SSID** field, type: **LabWLAN**.
4. Under the **Authentication** section, select the radio button for **WEP**.
5. In the **Key** field, enter the 10-digit hexadecimal key: **1234567890**.
6. Close the window; the device will automatically search for the Access Point and show wireless "dotted lines" to indicate a successful connection.

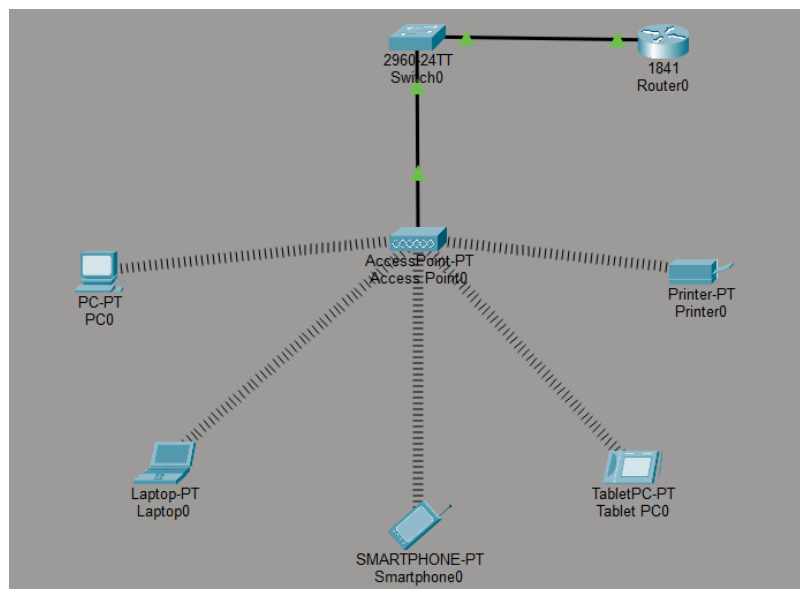
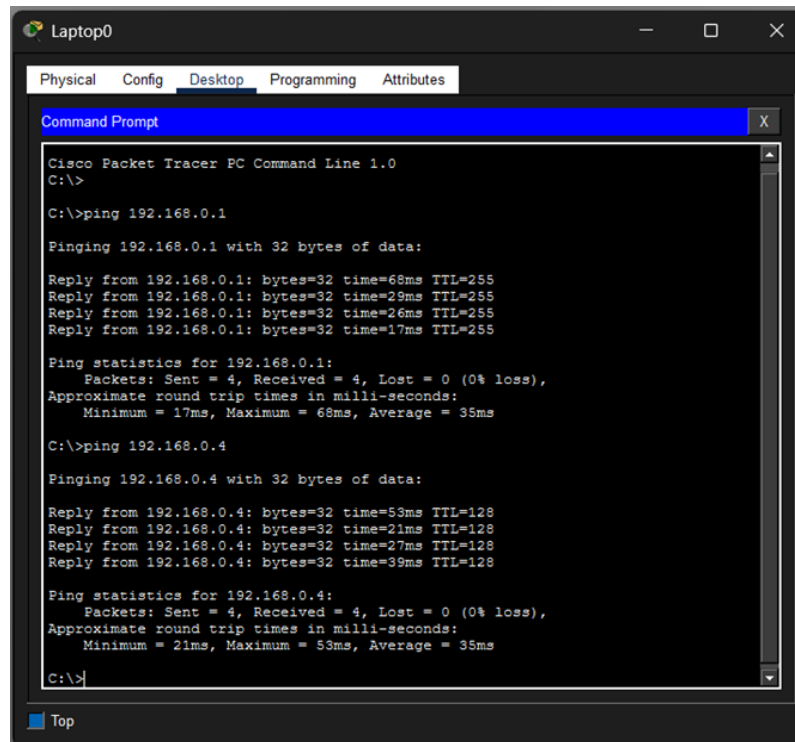


Figure 4: Wireless Device Connected to Access Point

- **Step 8: Verify Network Connectivity**

- Open **Command Prompt** on any device.
- Run the ping command:
 - ping 192.168.0.1 or ping 192.168.0.3
 -

Successful replies confirm that the network is functioning properly.



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Cisco Packet Tracer PC Command Line 1.0
C:\>

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=68ms TTL=255
Reply from 192.168.0.1: bytes=32 time=29ms TTL=255
Reply from 192.168.0.1: bytes=32 time=26ms TTL=255
Reply from 192.168.0.1: bytes=32 time=17ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 68ms, Average = 35ms

C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time=53ms TTL=128
Reply from 192.168.0.4: bytes=32 time=21ms TTL=128
Reply from 192.168.0.4: bytes=32 time=27ms TTL=128
Reply from 192.168.0.4: bytes=32 time=39ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 53ms, Average = 35ms

C:\>
  
```

Figure 5: Ping Test Showing Successful Connectivity

➤ Output:

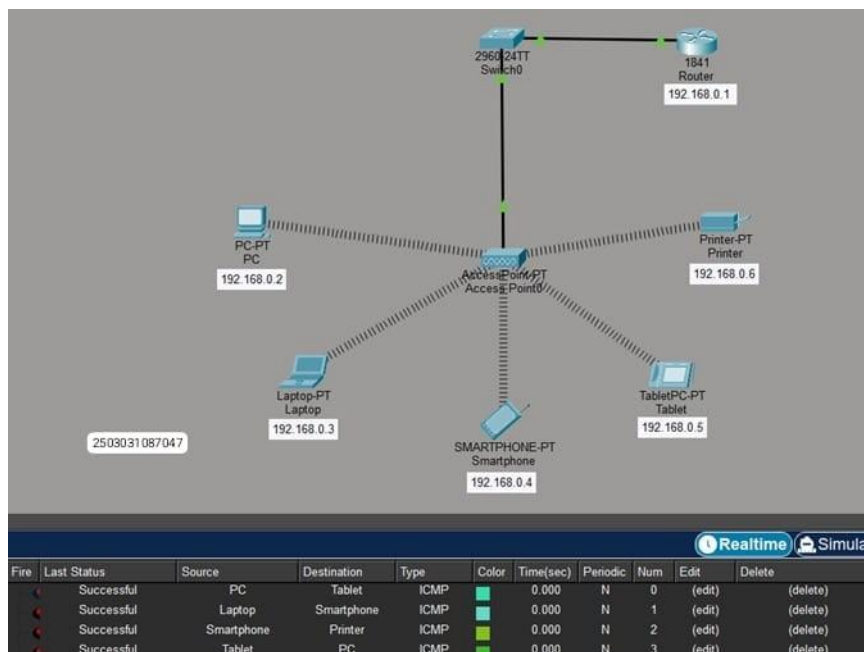


Figure 6: Wireless LAN (WLAN) topology showing wireless devices connected to an Access Point with successful packet transmission in Cisco Packet Tracer.



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➤ **OBSERVATION:**

- **Module Installation:** PCs and printers required manual hardware module swaps to enable wireless capabilities, whereas the smartphone and tablet connected natively.
- **Wireless Connectivity:** Once the SSIDs matched, the end devices successfully established a wireless link with the central Access Point.

Network Communication: Pings from wireless devices (e.g., Laptop at 192.168.0.3) to the wired Router gateway (192.168.0.1) were successful, confirming the Access Point effectively bridged the two network mediums.

➤ **CONCLUSION**

The experiment successfully demonstrates the configuration and design of a Wireless LAN (WLAN) using Cisco Packet Tracer. By deploying an Access Point and appropriately configuring end devices, it was observed that wireless clients can securely and effectively communicate with a central wired network. This practical highlights the fundamental steps required to integrate wireless mobility into traditional wired network infrastructures.

Practical-6

AIM: Design three or four simple networks (with 3 to 4 hosts) and connect via Router. Perform simulation and trace how routing is done in packet transmission.

A: Experiment on the same subnet.

B: Experiment across the subnet and observe the functioning of Router.

➤ REQUIREMENTS

1. Any Computer or Laptop
2. Computer system with minimum 4 GB RAM
3. Cisco Packet Tracer software
4. Windows Operating System

➤ THEORY

This experiment utilizes Variable Length Subnet Masking (VLSM) with a /26 mask (255.255.255.192), which creates subnets of 64 addresses. To connect these subnets, Static Routing is required.

- Intra-Subnet: Communication within the same network block (e.g., PC0 to PC1).
- Inter-Subnet: Communication between different network blocks (e.g., PC1 to PC2) requiring a "Next Hop" path defined in the router's routing table.

➤ PROCEDURE

- **Step 1: Network Topology Setup**
 - Drag and drop two 2911 Routers (Router0 and Router1).
 - Add two 2960 Switches and four PCs.
 - Connect PCs to Switches using Copper Straight-Through cables.
 - Connect Switches to Router Gigabit ports (Gig0/1).
 - Connect Router0 to Router1 using a Serial DCE or Cross-over cable (Gig0/0 to Gig0/0) to form the WAN link (Subnet 3).
- **Step 2: IP Configuration (Subnetting)**

Configure all devices using the /26 mask (255.255.255.192):



- Subnet 1 : Assign PC0 (.1), PC1 (.2), and Router0 Gateway (.3).
- Subnet 2 : Assign PC2 (.65), PC3 (.66), and Router1 Gateway (.67).
- Subnet 3 : Assign Router0 interface (.129) and Router1 interface (.130).

- **Step 3: Router CLI Configuration**

On Router0: Configure Gig0/1 as 192.168.10.3 and Gig0/0 as 192.168.10.129.

On Router1: Configure Gig0/1 as 192.168.10.67 and Gig0/0 as 192.168.10.130.

- **Step 4: Configuring Static Routes**

To allow traffic to move between the networks, add the following paths:

- Router0: ip route 192.168.10.64 via 192.168.10.130
- Router1: ip route 192.168.10.0 via 192.168.10.129

- **Step 5: Simulation and Tracing**

- Switch to Simulation Mode.
- Create a PDU from PC1 to PC2.
- Click Capture/Forward and observe the packet move from PC1 → Switch0 → Router0 → Router1 → Switch1 → PC2.
- Verify the "Successful" status in the Realtime tab.

➤ OUTPUT

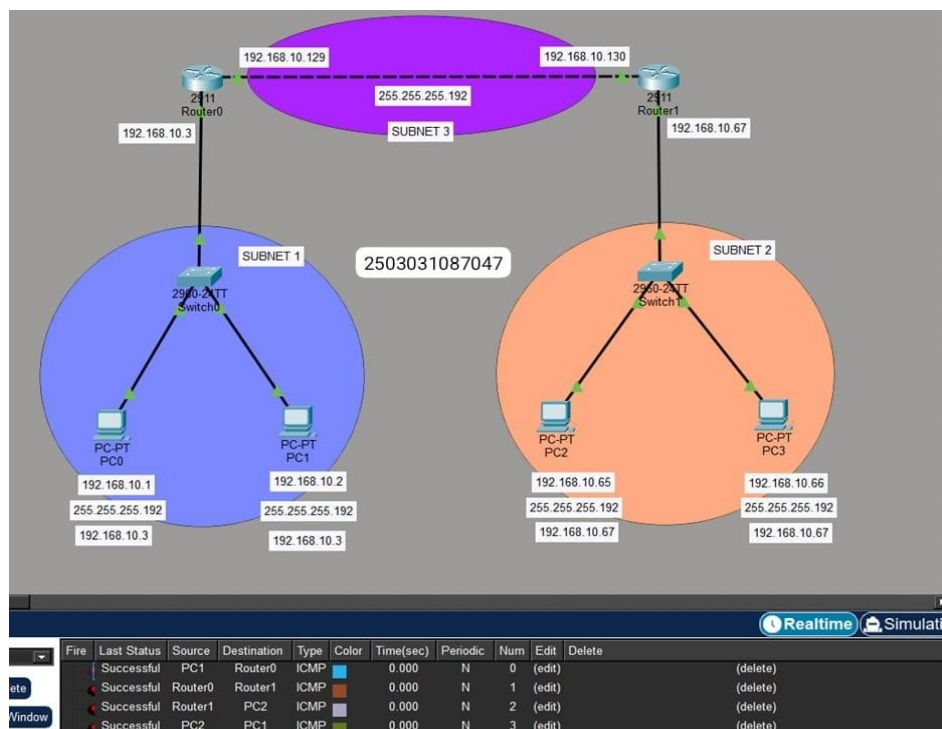


Figure 1: Implementation of Static Routing across three subnets using two 2911 Routers (Subnet Mask: 255.255.255.192).

➤ OBSERVATION

- Experiment A (Same Subnet): PC0 and PC1 were processed by the switch immediately. The ARP table was populated without router intervention.
- Experiment B (Across Subnet): PC1 to PC2 successfully traversed three subnets. The router used the Static Route to identify that the destination network 192.168.10.64 was reachable via the next hop 192.168.10.130.
- Packet Tracing: In simulation, it was observed that the Layer 2 header (MAC address) was rewritten at every router hop, while the Layer 3 header (IP address) remained constant.

➤ CONCLUSION

The design of three subnets connected via two routers was successfully implemented. Through Experiment A and B, we confirmed that routers are necessary for inter-subnet communication and that Static Routing is an effective way to manually define the data path. The simulation successfully traced the packet flow, demonstrating the hop-by-hop nature of network transmission.